BCS-089	Distributed System	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To introduce fundamental principles of distributed systems, technical challenges and key design issues.
CO2	To impart knowledge of the distributed computing models, algorithms and the design of distributed system
CO3	To understand the concept of deadlock
CO4	To understand the Mechanism for building distributed file systems
CO5	To understand concept of Distributed Mutual Exclusion

Detailed Syllabus

MODULE-I

Characterization of Distributed Systems: Introduction, Examples of distributed Systems, Resource sharing and the Web Challenges. Architectural models, Fundamental Models. **Theoretical Foundation for Distributed System**: Limitation of Distributed system, absence of global clock, shared memory, Lamport's Logical clock, Vectors clocks.

Concepts in Message Passing Systems: causal order, total order, Techniques for Message Ordering, Causal ordering of messages, global state, termination detection.

Distributed Mutual Exclusion: Classification of distributed mutual exclusion, requirement of mutual exclusion theorem, Token based and non token based algorithms, performance metric for distributed mutual exclusion algorithms.

MODULE-II

Distributed Deadlock Detection: system model, resource Vs communication deadlocks, deadlock prevention, avoidance, detection & resolution, centralized dead lock detection, distributed dead lock detection,path pushing algorithms, edge chasing algorithms. **Agreement Protocols**: Introduction, System models,classification of Agreement Problem, Byzantine agreement problem, Consensus problem, Interactive consistency Problem, Solution to Byzantine Agreement problem, Application of Agreement problem, Atomic Commit in Distributed Database system.

Distributed Resource Management: Issues in distributed File Systems, Mechanism for building distributed file systems, Design issues in Distributed Shared Memory, Algorithm for Implementation of Distributed Shared Memory**.**

MODULE-III

Failure Recovery in Distributed Systems: Concepts in Backward and Forward recovery, Recovery in Concurrent systems, obtaining consistent Checkpoints, Recovery in Distributed Database Systems.

Fault Tolerance: Issues in Fault Tolerance, Commit Protocols, Voting protocols, Dynamic voting protocols.

Transactions and Concurrency Control: Transactions, Nested transactions, Locks, Optimistic Concurrency control, Timestamp ordering.

Distributed Transactions: Flat and nested distributed transactions, Atomic Commit protocols, Concurrency control in distributed transactions, Distributed deadlocks, Transaction recovery. Replication: System model and group communication, Fault - tolerant services, highly available services, Transactions with replicated data.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Illustrate the mechanisms of inter process communication in distributed system
CO2	Compare the concurrency control mechanisms in distributed transactional environment
CO3	Outline the need for mutual exclusion and election algorithms in distributed systems
CO4	Can apply the concept of distributed transaction
CO5	Can apply the concept of Dynamic voting protocols

Text Books:

1. Singhal&Shivaratri, "Advanced Concept in Operating Systems", McGraw Hill
2. Coulouris, Dollimore, Kindberg, "Distributed System: Concepts and Design", Pearson Education

Reference Books:

1. Tenanuanbaum, Steen," Distributed Systems", PHI
2. Gerald Tel, "Distributed Algorithms", Cambridge University Press.